

ACG Adversarial & Penetration Testing Report

19 Live HTTP Vectors & 77 Automated Tests

1. Executive Testing Summary

* Automated Unit & Integration Tests: 77 / 77 Passing across 9 test files.

* Live Penetration Tests (npm run pentest): 19 / 19 Vectors Passed (100% Intercepted).

* Unauthorized Financial-Impact Paths Observed: 0.

2. Live HTTP Penetration Test Matrix

Test ID	Category	Attack Scenario	Expected HTTP	Observed HTTP	Financial Effect	Audit Status	Result
AUTH-01	Cryptographic	Valid Ed25519 Mandate Signature	201	201	Authorized	MANDATE_VERIFIED	PASS
AUTH-02	Cryptographic	Tampered Mandate Budget	401	401	None	SIGNATURE_VERIFICATION_FAILED	PASS
AUTH-03	Temporal	Expired Buyer Mandate	403	403	None	POLICY_VIOLATION	PASS
LOGIC-01	Commerce Truth	Hallucinated Price (DB Price Wins)	201	201	Authorized (DB total)	COMMERCE_TRUTH_RESOLVED	PASS
LOGIC-02	Policy	Mandate Budget Limit Exceeded	403	403	None	POLICY_VIOLATION	PASS
LOGIC-03	Policy	Category Not Permitted in Mandate	403	403	None	POLICY_VIOLATION	PASS
CONCUR-01	Concurrency	10 Parallel Subagents Competing	1x201, 9x409	1x201, 9x409	1 Order Created	DUAL_RESERVATION_HELD	PASS
REPLAY-01	Replay Gate	Duplicate intent_id Submission	409	409	None	Blocked at Session Gate	PASS
WEBHOOK-01	Webhook	Forged HMAC SHA-256 Signature	401	401	None	Rejected before DB read	PASS
REFUND-01	Refund Safety	Pre-Capture Refund Attempt	200 (Blocked)	200 (Blocked)	None	State remained ORDER_CREATED	PASS
POL-01	Dynamic Policy	Real-Time Policy Update to v2.0.0	200	200	None	Policy pinned to pol_v2.0.0	PASS
POL-02	Dynamic Policy	Over-Cap Checkout vs Mutated Policy	403	403	None	POLICY_VIOLATION	PASS
REV-01	Revocation	Principal Issues Mandate Revocation	200	200	None	Revocation logged in DB	PASS
REV-02	Revocation	Rogue Agent on Revoked Mandate	403	403	None	MANDATE_REVOKED	PASS
INPUT-01	Sanitization	Negative Item Quantity (-5)	400	400	None	Zod Schema Rejection	PASS
INPUT-02	Sanitization	SQL Injection in SKU Parameter	400	400	None	COMMERCE_TRUTH_REJECTION	PASS
AUDIT-01	Provenance	Verify Valid Cryptographic Hash Chain	200	200	None	96 blocks verified intact	PASS
AUDIT-02	Provenance	Detect Adversarial Record Tampering	200 (Tamper)	200 (Tamper)	None	Broken hash chain detected	PASS
AUTH-SCOPE	API Auth	Viewer Token on Privileged Route	403	403	None	Blocked at Bearer Scope Gate	PASS

